

Estimating Gender Discrimination in the Workplace

In this exercise we'll use data from the 2018 US Current Population Survey (CPS) to try and estimate the effect of being a woman on workplace compensation.

Note that our focus will be *only* on differential compensation in the work place, and as a result it is important to bear in mind that our estimates are not estimates of *all* forms of gender discrimination. For example, these analyses will not account for things like gender discrimination in terms of *getting* jobs. We'll discuss this in more detail below.

Exercise 1:

Begin by downloading and importing 2018 CPS data from https://github.com/nickeubank/MIDS_Data/tree/master/Current_Population_Survey. The file is called `morg18.dta` and is a Stata dataset. Additional data on the dataset can be found by following the links in the README.txt file in the folder, but for the moment it is sufficient to know this is a national survey run in the United States.

The survey does include some survey weights we won't be using (i.e. not everyone in the sample was included with the same probability), so the numbers we estimate will not be perfect estimates of the gender wage gap in the United States, but they are pretty close.

```
In [1]: import pandas as pd
import numpy as np

url = "https://github.com/nickeubank/MIDS_Data/raw/refs/heads/master/Current_Population_
df = pd.read_stata(url)

df.head()
```

Out[1]:	hhid	intmonth	hurespli	hrhrttype	minsamp	hrlonglk	hrsampl	hrhhid2	series
0	000004795110719	January	1.0	Husband/wife primary fam (neither in Armed For...	MIS 8	MIS 2-4 Or MIS 6-8 (link To	0601	06011	
1	000004795110719	January	1.0	Husband/wife primary fam (neither in Armed For...	MIS 8	MIS 2-4 Or MIS 6-8 (link To	0601	06011	
2	000110339935453	January	1.0	Unmarried civilian female primary fam householder	MIS 4	MIS 2-4 Or MIS 6-8 (link To	0701	07011	
3	000110339935453	January	1.0	Unmarried civilian female primary fam householder	MIS 4	MIS 2-4 Or MIS 6-8 (link To	0701	07011	
4	000110359424339	January	1.0	Unmarried civilian female primary fam householder	MIS 4	MIS 2-4 Or MIS 6-8 (link To	0711	07111	

5 rows x 98 columns

Exercise 2:

Because our interest is only in-the-workplace wage discrimination among full-time workers, we need to start by subsetting our data for people currently employed (and "at work", not "absent") at the time of this survey using the `lfsr94` variable, who are employed full time (meaning that their usual hours per week—`uhouse`—is 35 or above).

As noted above, this analysis will miss many forms of gender discrimination. For example, in dropping anyone who isn't working, we immediately lose any women who couldn't get jobs, or who chose to lose the workforce because the wages they were offered (which were likely lower than those offered men) were lower than they were willing / could accept. And in focusing on full time employees, we miss the fact women may not be offered full time jobs at the same rate as men.

```
In [6]: print(df["lfsr94"].value_counts(normalize=True))
print()
print(df["uhouse"].describe())
```

```
lfsr94
Employed-At Work      0.572584
Retired-Not In Labor Force  0.203157
Other-Not In Labor Force  0.123776
Disabled-Not In Labor Force  0.056641
Employed-Absent      0.021402
Unemployed-Looking    0.018967
Unemployed-On Layoff   0.003474
Name: proportion, dtype: float64
```

```
count    180100.000000
mean      35.864259
std       15.080222
min       -4.000000
25%       35.000000
50%       40.000000
75%       40.000000
max       99.000000
Name: uhourse, dtype: float64
```

```
In [ ]: df_fulltime = df[
        (df["lfsr94"].astype(str).str.contains("At Work", case=False, na=False))
        & (df["uhourse"] >= 35)
    ].copy()

print(f"Original dataset size: {len(df)}")
print(f"Filtered full-time dataset size: {len(df_fulltime)}")
```

```
Original dataset size: 302332
Filtered full-time dataset size: 133814
```

Exercise 3

Now let's estimate the basic wage gap for the United States!

Earnings per week worked can be found in the `earnwke` variable. Using the variable `sex` (1=Male, 2=Female), estimate the gender wage gap in terms of wages per hour worked!

(You may also find it helpful, for context, to estimate the average hourly pay by dividing weekly pay by `uhourse`.)

```
In [18]: # Calculate hourly wage
df_fulltime["hourly_wage"] = df_fulltime["earnwke"] / df_fulltime["uhourse"]

# Calculate average hourly wage by sex
hourly_gap = df_fulltime.groupby("sex")["hourly_wage"].mean()
print(hourly_gap)

# Calculate the absolute gap
male_hourly = hourly_gap.loc[1]
female_hourly = hourly_gap.loc[2]
absolute_hourly_gap = male_hourly - female_hourly

print()
print(f"Absolute hourly wage gap: ${absolute_hourly_gap:.2f}")
```

```
sex
1    27.883330
2    23.803158
Name: hourly_wage, dtype: float64
```

Absolute hourly wage gap: \$4.08

Exercise 4

Assuming 48 work weeks in a year, calculate annual earnings for men and women. Report the difference in dollars and in percentage terms.

```
In [19]: # Calculate annual earnings (48 weeks)
df_fulltime["annual_wage"] = df_fulltime["earnwke"] * 48

# Calculate average annual wage by sex
annual_gap = df_fulltime.groupby("sex")["annual_wage"].mean()
print(annual_gap)

# Calculate the absolute gap
male_annual = annual_gap.loc[1]
female_annual = annual_gap.loc[2]

diff_dollars = male_annual - female_annual
diff_percent = (diff_dollars / male_annual) * 100

print()
print(f"Absolute difference: ${diff_dollars: .2f}")
print(f"Percentage difference: {diff_percent: .2f}%")
```

```
sex
1    57827.223276
2    47312.787581
Name: annual_wage, dtype: float64
```

Absolute difference: \$ 10514.44
Percentage difference: 18.18%

Exercise 5

We just compared all full-time working men to all full-time working women. For this to be an accurate *causal* estimate of the effect of being a woman in the work place, what must be true of these two groups? What is one reason that this may *not* be true?

- For this to be an accurate causal estimate of the effect of being a woman in the workplace, what must be true of these two groups?
 - The two groups (men and women) must be perfectly exchangeable (all else being equal). This means that if the women in the dataset were turned into men, they would earn exactly what the men are earning, and vice versa. They must be identical in absolutely every characteristic that

affects wages (like education, years of experience, age, and choice of industry) except for their gender.

- What is one reason that this may not be true?
 - There are many structural differences between the populations of working men and working women that act as confounders (Omitted Variable Bias). For example, working women in the dataset may be younger on average, or there may be occupational sorting (men and women working in systematically different industries) and differences in educational backgrounds.

Exercise 6

One answer to the second part of Exercise 5 is that working women are likely to be younger, since a larger portion of younger women are entering the workforce as compared to older generations.

To *control* for this difference, let's now regress annual earnings on gender, age, and age-squared (the relationship between age and income is generally non-linear). What is the implied average annual wage difference between women and men? Is it different from your raw estimate?

```
In [21]: import statsmodels.formula.api as smf

df_fulltime["age_sq"] = df_fulltime["age"] ** 2

model_ex6 = smf.ols("annual_wage ~ sex + age + age_sq", data=df_fulltime).fit()

print(model_ex6.summary())
```

OLS Regression Results

Dep. Variable:	annual_wage	R-squared:	0.060
Model:	OLS	Adj. R-squared:	0.060
Method:	Least Squares	F-statistic:	2625.
Date:	Sat, 21 Feb 2026	Prob (F-statistic):	0.00
Time:	11:33:32	Log-Likelihood:	-1.4441e+06
No. Observations:	122603	AIC:	2.888e+06
Df Residuals:	122599	BIC:	2.888e+06
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4.895e+04	401.294	121.981	0.000	4.82e+04	4.97e+04
sex	-1.062e+04	181.123	-58.641	0.000	-1.1e+04	-1.03e+04
age	456.6349	6.839	66.766	0.000	443.230	470.040
age_sq	-0.2740	1.241	-0.221	0.825	-2.706	2.158

Omnibus:	18297.323	Durbin-Watson:	1.733
Prob(Omnibus):	0.000	Jarque-Bera (JB):	27620.546
Skew:	1.110	Prob(JB):	0.00
Kurtosis:	3.692	Cond. No.	353.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- Implied average annual wage difference: \$10,620.
- Is it different from the raw estimate? Yes, it is slightly higher than the raw unconditional gap of \$10,514.44. Controlling for age actually widened the gap slightly. This makes sense: if working women are younger on average, and younger people generally earn less, comparing men and women at the exact same age reveals a slightly larger underlying penalty for being female.

Exercise 7

In running this regression and interpreting the coefficient on `female`, what is the implicit comparison you are making? In other words, when we run this regression and interpreting the coefficient on `female`, we're basically pretending we are comparing two groups and assuming they are counterfactuals for one another. What are these two groups?

- When we add a control variable like age to a regression, it fundamentally changes what the coefficient represents. We are no longer comparing "all working men" to "all working women."
- Instead, the implicit comparison we are making is between "a working man and a working woman who are the exact same age". We are assuming that, holding age constant, these two groups act as counterfactuals for one another.

Exercise 8

Now let's add to our regression an indicator variable for whether the respondent has at least graduated high school, and an indicator for whether the respondent at least has a BA.

In answering this question, use the following table of codes for the variable `grade92`.

Education is coded as follows:

 CPS Educ Codes

```
In [ ]: # Create indicator for High School or more
df_fulltime["hs_or_more"] = (df_fulltime["grade92"] >= 39).astype(int)

# Create indicator for BA or more
df_fulltime["ba_or_more"] = (df_fulltime["grade92"] >= 43).astype(int)

# Fit the model
model_ex8 = smf.ols(
    "annual_wage ~ sex + age + age_sq + hs_or_more + ba_or_more",
    data=df_fulltime,
).fit()

print(model_ex8.summary())
```

OLS Regression Results

Dep. Variable:	annual_wage	R-squared:	0.259
Model:	OLS	Adj. R-squared:	0.259
Method:	Least Squares	F-statistic:	8573.
Date:	Sun, 22 Feb 2026	Prob (F-statistic):	0.00
Time:	21:44:47	Log-Likelihood:	-1.4296e+06
No. Observations:	122603	AIC:	2.859e+06
Df Residuals:	122597	BIC:	2.859e+06
Df Model:	5		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.878e+04	468.339	61.458	0.000	2.79e+04	2.97e+04
sex	-1.3e+04	161.440	-80.496	0.000	-1.33e+04	-1.27e+04
age	445.9737	6.074	73.426	0.000	434.069	457.878
age_sq	-0.3958	1.102	-0.359	0.719	-2.555	1.763
hs_or_more	1.373e+04	344.678	39.840	0.000	1.31e+04	1.44e+04
ba_or_more	2.756e+04	167.116	164.937	0.000	2.72e+04	2.79e+04

Omnibus:	14257.428	Durbin-Watson:	1.849
Prob(Omnibus):	0.000	Jarque-Bera (JB):	20565.673
Skew:	0.890	Prob(JB):	0.00
Kurtosis:	3.926	Cond. No.	507.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Exercise 9

In running this regression and interpreting the coefficient on `female`, what is the implicit comparison you are making? In other words, when we run this regression and interpreting the coefficient on `female`, we are once more basically pretending we are comparing two groups and assuming they are counter-factuals for one another. What are these two groups?

- We are now comparing a working man and a working woman of the exact same age who also have the exact same level of educational attainment (specifically, whether or not they have a high school diploma, and whether or not they have a bachelor's degree).
- By holding both age and education constant, we are filtering out wage differences that are purely driven by one gender having more or less education than the other in this dataset.

Exercise 10

Given how the coefficient on `female` has changed between Exercise 6 and Exercise 8, what can you infer about the educational attainment of the women in your survey data (as compared to the educational attainment of men)?

- In Exercise 6 (controlling only for age), the penalty for being female was -\$10,620.
- In Exercise 8 (controlling for age and education), the penalty jumped to -\$13,000.
- Because the wage gap got worse for women when we forced the model to compare men and women with the exact same education level, it means that before we held education constant, women's education levels were actually masking some of the gap.
- We can infer that women in this full-time working sample are, on average, more highly educated than the men. Their higher educational attainment was pulling their average wages up, which partially offset the underlying gender penalty in the raw and age-adjusted estimates.

Exercise 11

What does that tell you about the *potential outcomes* of men and women before you added education as a control?

- It tells us that before adding education, the potential outcomes for women were systematically biased upward relative to a true "ceteris paribus" (all else equal) comparison. Because the women had a "positive" confounder (higher education), comparing their raw wages to men's raw wages made the wage gap look artificially smaller than it truly is for people with identical backgrounds.
- If we counterfactually equalized education across both groups, women would earn significantly less than they do in the unadjusted data.

Exercise 12

Finally, let's include *fixed effects* for the type of job held by each respondent.

Fixed effects are a method used when we have a nested data structure in which respondents belong to groups, and those groups may all be subject to different pressures. In this context, for example, we can add fixed effects for the industry of each respondent—since wages often vary across industries, controlling for industry is likely to improve our estimates. Use `ind02` to control for industry.

(Note that fixed effects are very similar in principle to hierarchical models. There are some differences [you will read about](#) for our next class, but they are designed to serve the same role, just with slightly different mechanics).

When we add fixed effects for groups like this, our interpretation of the other coefficients changes. Whereas in previous exercises we were trying to explain variation in men and women's wages *across all respondents*, we are now effectively comparing men and women's wages *within each employment sector*. Our coefficient on `female`, in other words, now tells us how much less (on average) we would expect a woman to be paid than a man *within the same industry*, not across all respondents.

(Note that running this regression will result in lots of coefficients popping up you don't care about. We'll introduce some more efficient methods for adding fixed effects that aren't so messy in a later class -- for now, you can ignore those coefficients!)

```
In [25]: model_ex12 = smf.ols(  
    "annual_wage ~ sex + age + age_sq + hs_or_more + ba_or_more + C(ind02)",  
    data=df_fulltime,  
).fit()  
  
print(  
    f"New gender wage gap (with industry fixed effects): {model_ex12.params['sex']:.2f}"  
)
```

New gender wage gap (with industry fixed effects): -10860.93

Exercise 13

Now that we've added industry fixed effects, what groups are we implicitly treated as counter-factuals for one another now?

- We are now implicitly comparing a working man and a working woman who are the exact same age, have the exact same level of educational attainment, and work within the exact same employment sector (industry).
- We have stripped away age, education, and occupational sorting to look purely at the wage difference between identical profiles doing the same type of work.

Exercise 14

What happened to your estimate of the gender wage gap when you added industry fixed effects? What does that tell you about the industries chosen by women as opposed to men?

When you're done, please come read [this discussion](#).

- What happened: The estimated gender wage gap shrank (became less severe). It went from a penalty of -13,000 dollars when controlling for just age and education to a penalty of approximately -10,861 dollars once we controlled for industry.
- What this tells us: Because forcing the model to compare men and women within the same industry reduces the gap, it means that cross-industry differences were previously making the gap look worse. This tells us that women in this sample are disproportionately concentrated in (or sorted into) lower-paying industries compared to men. When we isolate the comparison to within-industry only, the direct penalty for being female is smaller than the overall penalty across the entire economy.